TSA Drone Project Plan

Parts List

- **Flight Controller**: ESP32-wroom-32
- **Power System**: OVONIC 4S 14.8V 100C 550mAh LiPo Battery
- **Motors**: Readytosky RS2205 230kv Brushless Motor (CW/CCW pairs) - 4 units
- **ESC**: AERO SELFIE 45A 4in1 ESC (2-6S BL_S)
- **Sensors**: MPU 6050 (Gyro/Accelerometer)

Pre-Flight Checks

Safety Verification

- Battery voltage and connector security
- Motor propeller installation (CW/CCW pairing)
- RC transmitter binding and signal check
- Firmware version compatibility

Component Connection Validation

- All 4 motors connected to correct ESC ports
- Gyro/MPU 6050 properly wired to flight controller
- ESC-BLHeli_S firmware version check
- Power distribution board integrity

Boot Up Sequence

Mode 1: Set-Up / Troubleshoot Mode

Activated by holding both trigger buttons on startup.

1. **Gyro Initialization Check**
    - System powers up and reads MPU 6050 raw values
    - Expects ~0° pitch/roll when drone is level
    - If gyro does not return to 0 within 2 seconds: **flash red LED error pattern** and sound buzzer
    - Display: "GYRO ERROR - Check connections"
2. **Motor Detection & Direction Test**
    - System sequentially commands each motor:
        - **Front Right**: Should cause drone to tilt backward-left
        - **Front Left**: Should cause drone to tilt backward-right
        - **Back Left**: Should cause drone to tilt forward-right
        - **Back Right**: Should cause drone to tilt forward-left (default)
    - If any motor produces unexpected direction: **flash alternating red/green**, log error code
    - All motors spin up to 30% throttle, then slowly ramp down over 5 seconds

3. **Final Status**
    - Green LED solid: All checks passed, ready for normal mode
    - Red LED flash pattern: Specific failure mode identified

Mode 2: Normal Mode

Default boot mode if no triggers held.

1. **Startup Sequence**
    - Power-on self-test (POST): 200ms
    - Gyro calibration: 1 second (drone must remain stationary)
    - ESC startup beep sequence
    - RSSI signal strength display
2. **Ready State**
    - Solid green LED
    - Bluetooth/telemetry connection established
    - Ready for flight mode selection

Flight Modes

- **Manual**: Full stick control, no assistance
- **Stabilize**: Auto-leveling via gyro input
- **Position Hold**: GPS-assisted hovering (if module added)
- **Return-to-Launch**: Auto-return on signal loss

Post-Flight

- Battery level logging
- Motor heat check
- Firmware update check
- Log data review for tuning adjustments

Troubleshooting Guide

Symptom	Likely Cause	Action
Red LED on boot	Gyro misconnection	Check MPU 6050 wiring, re-seat connector
Motors not spinning	ESC not armed	Arm via throttle stick bottom-left sequence
Drift in stabilize mode	Gyro offset needs calibration	Re-run gyro calibration on level surface
One motor not responding	Motor/ESC failure	Swap motor ports to isolate fault
No RSSI signal	TX/RX mismatch	Check binding, antenna connection, frequency

Testing Protocol

1. **Bench Test**: All motors spin, direction verification, ESC beep tones
2. **Hover Test**: 30-second hover at 1m altitude, check stability
3. **Control Response**: Full pitch/roll/yaw movement in all directions
4. **Fail-Safe**: Signal loss triggers expected behavior (land or hover)